SHRI GOVINDRAM SEKSARIA INSTITUTE OF TECHNOLOGY AND SCIENCE INDORE (M.P)

Department of Computer Technology and Applications

MCA First Year

Semester - II

Session 2025-26

Subject: - Operating System

LAB ASSIGNMENT - II

Submitted To: -

Prof. Priyanka Kabra
Prof. Shweta Gupta

Submitted By:

Rishita Shah
(0801CA251116)

INDEX

S.No.	Assignment Questions	Page No.	Signature
1	**<u>Creating a file and a file name students.txt contains student records in the following format:</u>** RollNo:Name:Department:Marks 101:Rahul:CSE:78 102:Asha:IT:85 103:Ravi:CSE:65 104:Neha:ECE:90 105:Karan:CSE:78 Tasks : ♦ Display all records of CSE students. ♦ Count the total number of students. ♦ Display names of students who scored more than 75 marks. ♦ Sort the file based on marks (descending order). ♦ Find students having duplicate marks. **Using the grep, wc, awk, sort, uniq commands for this task**	1	

2	**<u>Case Study : Payroll Processing</u>** Creating an employee salary file name salary.txt and contains the following records in the following format: EmpID Name Department Salary E101 Amit HR 30000 E102 Rina CTA 45000 E103 Suresh CTA 50000 E104 Meena HR 35000	**4**	
	Perform the following task : ♦ Display employees belonging to the IT department. ♦ Print only Name and Salary columns. ♦ Increase salary by 10% for IT employees. ♦ Calculate the average salary. ♦ Count employees in each department. **Using the awk, cut, grep commands.**		
3	**<u>Case study :Automated Monitoring Report</u>** ♦ A daily monitoring report is required for system health. TasksCollect CPU, memory, and disk usage information. ♦ Save the report with date and time. ♦ Schedule the report generation automatically. ♦ View historical monitoring data. **Using top, free, df, date, cron, sar for complete these tasks.**	**7**	

4	**<u>Case Study :File Copy, Move, and Rename</u>** **Scenario** **Project files need to be organized into appropriate directories.** **<u>Tasks:</u>** ♦ Create a directory project. ♦ Copy report.txt into the project directory. ♦ Rename report.txt to final_report.txt. ♦ Move final_report.txt to a directory named backup. ♦ Verify the file location. **Using mkdir, cp, mv, ls commands for this task.**	**10**	
	<u>Case Study :File Deletion and Recovery</u>		
5	**<u>Awareness</u>** **Scenario** **Old and unused files must be removed from the system.** **Tasks:** ♦ Delete a file named old.txt. ♦ Remove an empty directory. ♦ Remove a directory containing files. ♦ List files before and after deletion. ♦ Explain why deleted files cannot be easily recovered. **Using rm, rmdir, ls for this task.**	**13**	

6	**<u>Check Current User and Group:</u>** Open a terminal and run the command to check your current user and group. **Setting Up the Environment:** ♦ Create a directory named file_permission_lab. ♦ Inside this directory, create three files: file1.txt, file2.txt, and file3.txt.	**15**	
7	**<u>Viewing File Permissions:</u>** ♦ Use the command to view the permissions of the files you created. Document the output.	**17**	
8	**<u>Changing File Permissions:</u>** ♦ Change the permissions of file1.txt to allow the owner to read and write, the group to read, and others to have no permissions. Use both symbolic and numeric methods. ♦ Change the permissions of file2.txt to allow everyone to read and execute but not write. ♦ Change the permissions of file3.txt to	**18**	
	allow only the owner to read, write, and execute.		
9	**<u>Documenting Changes:</u>** ♦ After each permission change, use ls -l to document the new permissions for each file.	**20**	

10	**<u>File Handling Tasks:</u>** **1.** Create a script called file_handling.sh that: · Creates a directory named test_dir. · Creates a file named test_file.txt inside test_dir. · Writes "This is a test file." to the file. · Displays the contents of the file. · Deletes the file and the directory. **2.** Make the script executable and run it with different arguments.	**21**	
11	**<u>Change Ownership:</u>** ♦ If you have another user account on the system, change the ownership of file1.txt to that user (replace other user with the actual username)	**22**	
12	**<u>Cleanup:</u>** ♦ After completing the tasks, remove the file_permission_lab directory and its contents	**23**	
13	**<u>File commands:</u>** **Create a Directory:** ♦ Create a new directory named file_commands_lab in your home directory. **Navigate to the Directory:** ♦ Change to the newly created directory ♦ Create five text files named file1.txt, file2.txt, file3.txt, file4.txt, and file5.txt.	**24**	
	♦ Use the command to list the files in the directory.		

14	**<u>Write to Files:</u>** ♦ Use the echo command to write “This is file 1” text into file1.txt on the terminal not in the directly writing text file. ♦ Append text to file2.txt.write the contents "This is file 2” in the file2.txt and that append.	**26**	
15	♦ Use the cat command to display the contents of file1.txt. ♦ Use the less command to view the contents of ‘file2.txt’	**27**	
16	**<u>Copy Files:</u>** ♦ Copy ‘file1.txt’ to create a new file named ‘file1_copy.txt’ ♦ Rename ‘file2.txt ‘ to ‘file2_renamed.txt’ ♦ Delete ‘file3.txt’. ♦ Change to the ‘backup directory and list the files:	**28**	
17	**<u>Create a Compressed Archive:</u>** ♦ Go back to the ‘file_commands_lab’ directory and create a compressed archive of the backup directory.	**30**	
18	**<u>Extract the Archive:</u>** ♦ Create a new directory named ‘extracted_backup’ and extract the contents of ‘backup.tar.gz’ into it:	**31**	
19	**<u>Verify Extraction:</u>** ♦ List the contents of the ‘extracted_backup’ directory to verify that the files were extracted correctly.	**32**	
20	**Create a text file named “ count.txt “ and write the 10 line sentences and use the command to count the word ,lines , character and also use command for highlighting a patten in this file .**	**33**	

Question 1:

<u>Creating a file and a file name students.txt contains student records in the following format:</u>

RollNo:Name:Department:Marks
101:Rahul:CSE:78
102:Asha:IT:85
103:Ravi:CSE:65
104:Neha:ECE:90
105:Karan:CSE:78

Tasks :

- **Display all records of CSE students.**
- **Count the total number of students.**
- **Display names of students who scored more than 75 marks.**
- **Sort the file based on marks (descending order).**
- **Find students having duplicate marks.**

Using the grep, wc, awk, sort, uniq commands for this task

<u>Commands:</u>

rishita@kali:~/rishita$ touch students.txt
rishita@kali:~/rishita$ open students.txt
rishita@kali:~/rishita$ awk -F ":" '{print}' students.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ touch students.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ open students.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ awk -F ":" '{print}' students.txt
RollNo:Name:Department:Marks
101:Rahul:CSE:78
102:Asha:IT:85
103:Ravi:CSE:65
104:Neha:ECE:90
105:Karan:CSE:78
```

1. **Display all records of CSE students.**

Command:
rishita@kali:~/rishita$ grep "CSE" students.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ grep "CSE" students.txt
101:Rahul:CSE:78
103:Ravi:CSE:65
105:Karan:CSE:78
```

2. **Count the total number of students.**

Command:
rishita@kali:~/rishita$ wc -l students.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ wc -l students.txt
6 students.txt
```

3. **Display names of students who scored more than 75 marks.**

Command:
rishita@kali:~/rishita$ awk -F: '$4 > 75 {print $2}' students.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ awk -F: '$4 > 75 {print $2}' students.txt
Rahul
Asha
Neha
Karan
```

4. **Sort the file based on marks (descending order).**

Command:
rishita@kali:~/rishita$ sort -t: -k4 -nr students.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ sort -t: -k4 -nr students.txt
104     : Neha  :       ECE             :       90
102     : Asha  :       IT              :       85
105     : Karan :       CSE             :       78
101     : Rahul :       CSE             :       78
103     : Ravi  :       CSE             :       65
RollNo  : Name  :       Department      :       Marks
```

5. **Find students having duplicate marks.**

Command:

rishita@kali:~/rishita$ awk -F: '{print $4 ":" $0}' students.txt | sort | uniq -d -w2 | cut -d: -f2-

```
┌──(rishita㉿kali)-[~/rishita]
└─$ awk -F: '{print $4 ":" $0}' students.txt | sort |
uniq -d -w2 | cut -d: -f2-
101     : Rahul :       CSE             :       78
```

Question 2:

<u>Case Study : Payroll Processing</u>

Creating an employee salary file name salary.txt and contains the following records in the following format:

EmpID Name Department Salary
E101 Amit HR 30000
E102 Rina CTA 45000
E103 Suresh CTA 50000
E104 Meena HR 35000

Perform the following task :

1. **Display employees belonging to the IT department.**
2. **Print only Name and Salary columns.**
3. **Increase salary by 10% for IT employees.**
4. **Calculate the average salary.**
5. **Count employees in each department.**

Using the awk, cut, grep commands.

Commands:

rishita@kali:~/rishita$ touch salary.csv

rishita@kali:~/rishita$ libreoffice salary.csv

	A	B	C	D
1	EmpID	Name	Department	Salary
2	E101	Amit	HR	30000
3	E102	Rina	CTA	45000
4	E103	Suresh	CTA	50000
5	E104	Meena	HR	35000
6	E105	Ram	IT	55000
7	E106	Shyam	IT	60000

1. **Display employees belonging to the IT department.**

Command:

rishita@kali:~/rishita$ grep "IT" salary.csv

```
E106,Shyam,IT,66000
```

2. **Print only Name and Salary columns.**

Command:

rishita@kali:~/rishita$ awk -F, '{print $2,$4}' salary.csv

```
Shyam 60000
```

3. **Increase salary by 10% for IT employees.**

Command:

rishita@kali:~/rishita$ awk -F, '{if ($3=="IT") $4 = $4 * 1.10; print}' salary.csv

```
┌──(rishita㉿kali)-[~/rishita]
└─$ awk -F, '{if ($3=="IT") $4 = $4 * 1.10; print}' salary.csv
EmpID,Name,Department,Salary
E101,Amit,HR,30000
E102,Rina,CTA,45000
E103,Suresh,CTA,50000
E104,Meena,HR,35000
E105 Ram IT 60500
E106 Shyam IT 66000
```

4. **Calculate the average salary.**

Command:

rishita@kali:~/rishita$ awk -F, 'NR>1 { sum += $4; count++ } END { print "Average Salary =", sum/count }' salary.csv

```
┌──(rishita㉿kali)-[~/rishita]
└─$ awk -F, 'NR>1 { sum += $4; count++ } END { print "Average Salary =",
Average Salary = 45833.3
```

5. **Count employees in each department.**

Command:

```
rishita@kali:~/rishita$ awk -F, 'NR>1 { dept[$3]++ } END {for (d in dept) print d, dept[d]}'
salary.csv
```

Question 3:

<u>Case study: Automated Monitoring Report</u>

A daily monitoring report is required for system health.

Tasks:

1. **Collect CPU, memory, and disk usage information.**
2. **Save the report with date and time.**
3. **Schedule the report generation automatically.**
4. **View historical monitoring data.**

Using top, free, df, date, cron, sar for complete these tasks.

1. **Collect CPU, memory, and disk usage information.**

Commands:
rishita@kali:~/rishita$ top -b -n 1 | head -5
rishita@kali:~/rishita$ free -h
rishita@kali:~/rishita$ df -h

```
┌──(rishita㉿kali)-[~/rishita]
└─$ top -b -n 1 | head -5
top - 14:51:21 up 10 min,  1 user,  load average: 0.83, 0.63, 0.38
Tasks: 399 total,   1 running, 398 sleeping,   0 stopped,   0 zombie
%Cpu(s):  2.5 us,  2.5 sy,  0.0 ni, 94.5 id,  0.5 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  15666.7 total,  10386.5 free,   3013.8 used,   2999.2 buff/cache
MiB Swap:   7926.0 total,   7926.0 free,      0.0 used.  12653.0 avail Mem

┌──(rishita㉿kali)-[~/rishita]
└─$ free -h
               total        used        free      shared  buff/cache   available
Mem:            15Gi       3.0Gi        10Gi       412Mi       3.0Gi        12Gi
Swap:          7.7Gi          0B       7.7Gi

┌──(rishita㉿kali)-[~/rishita]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            7.7G     0  7.7G   0% /dev
tmpfs           1.6G  1.9M  1.6G   1% /run
/dev/nvme0n1p7  139G  100G   31G  77% /
tmpfs           7.7G  4.0K  7.7G   1% /dev/shm
efivarfs        256K  160K   92K  64% /sys/firmware/efi/efivars
tmpfs           5.0M     0  5.0M   0% /run/lock
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           7.7G  8.0K  7.7G   1% /tmp
/dev/nvme0n1p6  976M  160K  975M   1% /boot/efi
tmpfs           1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           1.6G  124K  1.6G   1% /run/user/1000
```

2. **Save the report with date and time.**

Commands:
rishita@kali:~/rishita$ date > report.txt
rishita@kali:~/rishita$ open report.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ date > report.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ open report.txt
```

```
Terminal
File  Edit  View  Search  Terminal  Help
Sunday 08 March 2026 02:53:32 PM IST
~
~
```

3. **Schedule the report generation automatically.**

Commands: rishita@kali:~/rishita$ crontab -e
rishita@kali:~/rishita$ crontab -l
rishita@kali:~/rishita$ cat cron_test.log

```
┌──(rishita㉿kali)-[~/rishita]
└─$ crontab -e
crontab: installing new crontab

┌──(rishita㉿kali)-[~/rishita]
└─$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command

* * * *  * echo "Cron is working" >> /home/cron_test.log

┌──(rishita㉿kali)-[~/rishita]
└─$ cat ~/cron_test.log
Cron is working
Cron is working
```

4. **View historical monitoring data.**

Commands:
rishita@kali:~/rishita$ sar

```
┌──(rishita㉿kali)-[~/rishita]
└─$ sar
Linux 6.12.33+kali-amd64 (kali)         08/03/26        _x86_64_        (16 CPU)

03:24:03 PM IST  LINUX RESTART  (16 CPU)
```

Question 4:

<u>Case Study :File Copy, Move, and Rename</u>

Scenario

Project files need to be organized into appropriate directories.

<u>Tasks:</u>

1. **Create a directory project.**
2. **Copy report.txt into the project directory.**
3. **Rename report.txt to final_report.txt.**
4. **Move final_report.txt to a directory named backup.**
5. **Verify the file location.**

Using mkdir, cp, mv, ls commands for this task.

1. **Create a directory project.**

Commands:

rishita@kali:~/rishita$ mkdir project

```
┌──(rishita㉿kali)-[~/rishita]
└─$ mkdir project
```

2. **Copy report.txt into the project directory.**

Commands:

rishita@kali:~/rishita$ cp report.txt project/

rishita@kali:~/rishita$ ls project/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ cp report.txt project/

┌──(rishita㉿kali)-[~/rishita]
└─$ ls project/
report.txt
```

3. **Rename report.txt to final_report.txt.**

Commands:

rishita@kali:~/rishita$ mv project/report.txt project/final_report.txt

rishita@kali:~/rishita$ ls project/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ mv project/report.txt project/final_report.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ ls project/
final_report.txt
```

4. **Move final_report.txt to a directory named backup.**

Commands:

rishita@kali:~/rishita$ mkdir backup

rishita@kali:~/rishita$ mv project/final_report.txt backup/

rishita@kali:~/rishita$ ls backup/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ mkdir backup

┌──(rishita㉿kali)-[~/rishita]
└─$ mv project/final_report.txt backup/
```

5. **Verify the file location.**
6.

Commands:

rishita@kali:~/rishita$ ls backup/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ ls backup/
final_report.txt
```

Question 5:

<u>Case Study :File Deletion and Recovery Awareness</u>

<u>Scenario</u>
<u>Old and unused files must be removed from the system.</u>

<u>Tasks:</u>

1. **Delete a file named old.txt.**
2. **Remove an empty directory.**
3. **Remove a directory containing files.**
4. **List files before and after deletion.**
5. **Explain why deleted files cannot be easily recovered.**

<u>Using rm, rmdir, ls for this task.</u>

1. **Delete a file named old.txt.**

Commands: rishita@kali:~/rishita$ touch old.txt
rishita@kali:~/rishita$ rm old.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  old.txt  project  report.txt  salary.csv  salary.txt  students.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ rm old.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  project  report.txt  salary.csv  salary.txt  students.txt
```

2. **Remove an empty directory.**

Commands:
rishita@kali:~/rishita$ rmdir empty/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ mkdir empty

┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  empty  project  report.txt  salary.csv  salary.txt  students.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ rmdir empty/

┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  project  report.txt  salary.csv  salary.txt  students.txt
```

3. **Remove a directory containing files.**

Commands:
rishita@kali:~/rishita$ rm -r file/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ mkdir file

┌──(rishita㉿kali)-[~/rishita]
└─$ mkdir file/new1

┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  file  project  report.txt  salary.csv  salary.txt  students.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ ls file/
new1

┌──(rishita㉿kali)-[~/rishita]
└─$ touch file/new.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ ls file/
new1  new.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ rm -r file/

┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  project  report.txt  salary.csv  salary.txt  students.txt
```

4. **List files before and after deletion.**

Commands: rishita@kali:~/rishita$ ls
rishita@kali:~/rishita$ ls file/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  file  project  report.txt  salary.csv  salary.txt  students.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ touch file/new1.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ ls file/
new1  new1.txt  new.txt

┌──(rishita㉿kali)-[~/rishita]
└─$ rm -r file/

┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  project  report.txt  salary.csv  salary.txt  students.txt
```

5. **Explain why deleted files cannot be easily recovered.**

Explanation:
When a file is deleted using rm, the system removes its reference from the file system. The actual data may be overwritten later, making recovery difficult. Deleted files cannot be easily recovered primarily because when you delete a file, the operating system does not actually remove the data from the storage device. Instead, it deletes the file's index (its entry in the file system table) and marks the space it occupied as "available" for new data.

Question 6:

<u>Check Current User and Group:</u>
Open a terminal and run the command to check your current user and group.

Setting Up the Environment:

1. **Create a directory named file_permission_lab.**
2. **Inside this directory, create three files: file1.txt, file2.txt, and file3.txt.**

Commands: rishita@kali:~/rishita$ whoami
rishita@kali:~/rishita$ groups

```
┌──(rishita㉿kali)-[~/rishita]
└─$ whoami
rishita

┌──(rishita㉿kali)-[~/rishita]
└─$ groups
rishita adm dialout cdrom floppy sudo audio dip www-data video plugdev users
hark kaboxer
```

1. **Create a directory named file_permission_lab.**

Commands:
rishita@kali:~/rishita$ mkdir file_permission_lab

```
┌──(rishita㉿kali)-[~/rishita]
└─$ mkdir file_permission_lab

┌──(rishita㉿kali)-[~/rishita]
└─$ ls
backup  file_permission_lab  project  report.txt  salary.csv  salary.txt  students.txt
```

2. **Inside this directory, create three files: file1.txt, file2.txt, and file3.txt.**

Commands:

rishita@kali:~/rishita$ cd file_permission_lab/

rishita@kali:~/rishita/file_permission_lab$ touch file1.txt file2.txt file3.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ cd file_permission_lab/

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ touch file1.txt file2.txt file3.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls
file1.txt  file2.txt  file3.txt
```

Question 7:

<u>Viewing File Permissions:</u>

- **Use the command to view the permissions of the files you created. Document the output.**

Commands:

rishita@kali:~/rishita$ ls -l

```
┌──(rishita㉿kali)-[~/rishita]
└─$ ls -l
total 28
drwxrwxr-x 2 rishita rishita 4096 Mar  8 15:27 backup
drwxrwxr-x 2 rishita rishita 4096 Mar  8 15:40 file_permission_lab
drwxrwxr-x 2 rishita rishita 4096 Mar  8 15:27 project
-rw-rw-r-- 1 rishita rishita   37 Mar  8 14:53 report.txt
-rw-rw-r-- 1 rishita rishita  110 Feb 25 21:14 salary.csv
-rw-rw-r-- 1 rishita rishita  110 Feb 25 21:30 salary.txt
-rw-rw-r-- 1 rishita rishita  152 Feb 25 20:47 students.txt
```

Question 8:

<u>Changing File Permissions:</u>

- **Change the permissions of file1.txt to allow the owner to read and write, the group to read, and others to have no permissions. Use both symbolic and numeric methods.**

Commands:

rishita@kali:~/rishita/file_permission_lab$ ls -l
rishita@kali:~/rishita/file_permission_lab$ chmod u=rw,g=r,o= file1.txt
rishita@kali:~/rishita/file_permission_lab$ ls –l

```
┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l
total 0
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file1.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file2.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file3.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ chmod u=rw,g=r,o= file1.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l
total 0
-rw-r----- 1 rishita rishita 0 Mar  8 15:40 file1.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file2.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file3.txt
```

- **Change the permissions of file2.txt to allow everyone to read and execute, but not write.**

Commands:

rishita@kali:~/rishita/file_permission_lab$ ls -l

rishita@kali:~/rishita/file_permission_lab$ chmod a=rx file2.txt

rishita@kali:~/rishita/file_permission_lab$ ls –l

```
┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l
total 0
-rw-r----- 1 rishita rishita 0 Mar  8 15:40 file1.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file2.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file3.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ chmod a=rx file2.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l
total 0
-rw-r----- 1 rishita rishita 0 Mar  8 15:40 file1.txt
-r-xr-xr-x 1 rishita rishita 0 Mar  8 15:40 file2.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file3.txt
```

- **Change the permissions of file3.txt to allow only the owner to read, write, and execute.**

Commands:

rishita@kali:~/rishita/file_permission_lab$ ls -l

rishita@kali:~/rishita/file_permission_lab$ chmod a=rx file2.txt

rishita@kali:~/rishita/file_permission_lab$ ls –l

```
┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l
total 0
-rw-r----- 1 rishita rishita 0 Mar  8 15:40 file1.txt
-r-xr-xr-x 1 rishita rishita 0 Mar  8 15:40 file2.txt
-rw-rw-r-- 1 rishita rishita 0 Mar  8 15:40 file3.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ chmod a=rx file3.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l
total 0
-rw-r----- 1 rishita rishita 0 Mar  8 15:40 file1.txt
-r-xr-xr-x 1 rishita rishita 0 Mar  8 15:40 file2.txt
-r-xr-xr-x 1 rishita rishita 0 Mar  8 15:40 file3.txt
```

Question 9:

<u>Documenting Changes:</u>

- **After each permission change, use ls -l to document the new permissions for each file.**

Commands:
rishita@kali:~/rishita/file_permission_lab$ ls -l

```
┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l
total 0
-rw-r----- 1 rishita rishita 0 Mar  8 15:40 file1.txt
-r-xr-xr-x 1 rishita rishita 0 Mar  8 15:40 file2.txt
-r-xr-xr-x 1 rishita rishita 0 Mar  8 15:40 file3.txt
```

Question 10:

File Handling Tasks:

1. Create a script called file_handling.sh that:

- **Creates a directory named test_dir.**
- **Creates a file named test_file.txt inside test_dir.**
- **Writes "This is a test file." to the file.**
- **Displays the contents of the file.**
- **Deletes the file and the directory.**

2. Make the script executable and run it with different arguments.

Commands:
rishita@kali:~/rishita$ touch file_handling.sh
rishita@kali:~/rishita$ vim file_handling.sh
rishita@kali:~/rishita$ chmod +x file_handling.sh
rishita@kali:~/rishita$ ./file_handling.sh
rishita@kali:~/rishita$./file_handling.sh test

```
┌──(rishita㉿kali)-[~]
└─$ touch file_handling.sh

┌──(rishita㉿kali)-[~]
└─$ vim file_handling.sh

┌──(rishita㉿kali)-[~]
└─$ chmod +x file_handling.sh

┌──(rishita㉿kali)-[~]
└─$ cat ./file_handling.sh
this is a test file.
```

Question 11:

<u>Change Ownership:</u>

- **If you have another user account on the system, change the ownership of file1.txt to that user (replace other user with the actual username)**

Commands:

rishita@kali:~/rishita$ cd file_permission_lab/

rishita@kali:~/rishita/file_permission_lab$ sudo chown linux file1.txt

rishita@kali:~/rishita/file_permission_lab$ ls -l file1.txt

```
┌──(rishita㉿kali)-[~/rishita]
└─$ cd file_permission_lab/

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ sudo chown rishita file1.txt

┌──(rishita㉿kali)-[~/rishita/file_permission_lab]
└─$ ls -l file1.txt
-rw-r——— 1 rishita rishita 0 Mar  8 15:40 file1.txt
```

Question 12:

<u>Cleanup:</u>

- **After completing the tasks, remove the file_permission_lab directory and its contents.**

Commands:

rishita@kali:~/rishita$ rm -r file_permission_lab/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ rm -r file_permission_lab/
rm: remove write-protected regular empty file 'file_permission_lab/file2.txt'? yes
```

Question 13:

<u>File commands:</u>

Create a Directory:

- **Create a new directory named file_commands_lab in your home directory.**

Commands:

rishita@kali:~/rishita$ mkdir file_commands_lab

```
┌──(rishita㉿kali)-[~/rishita]
└─$ mkdir file_commands_lab
```

Navigate to the Directory:

- **Change to the newly created directory**

Commands:

rishita@kali:~/rishita$ cd file_commands_lab/

```
┌──(rishita㉿kali)-[~/rishita]
└─$ cd file_commands_lab/

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$
```

- **Create five text files named file1.txt, file2.txt, file3.txt, file4.txt, and file5.txt.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ touch file1.txt file2.txt file3.txt file4.txt file5.txt

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
```

- **Use the command to list the files in the directory.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ ls

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.txt
```

Question 14:

<u>Write to Files:</u>

- **Use the echo command to write “This is file 1” text into file1.txt on the terminal not in the directly writing text file.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ echo "This is file 1" > file1.txt

rishita@kali:~/rishita/file_commands_lab$ cat file1.txt

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ echo "This is file 1" > file1.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ cat file1.txt
This is file 1
```

- **Append text to file2.txt.write the contents "This is file 2” in the file2.txt and that append.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ echo "This is file 2" >> file2.txt

rishita@kali:~/rishita/file_commands_lab$ cat file2.txt

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ echo "This is file 2" >> file2.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ cat file2.txt
This is file 2
```

Question 15:

- **Use the cat command to display the contents of file1.txt.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ cat file1.txt

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ cat file1.txt
This is file 1
```

- **Use the less command to view the contents of 'file2.txt'**

Commands:

rishita@kali:~/rishita/file_commands_lab$ cat file2.txt

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ cat file2.txt
This is file 2
```

Question 16:

<u>Copy Files:</u>

- **Copy 'file1.txt' to create a new file named 'file1_copy.txt'**

Commands:

rishita@kali:~/rishita/file_commands_lab$ cp file1.txt file1_copy.txt
rishita@kali:~/rishita/file_commands_lab$ ls

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ cp file1.txt file1_copy.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls
file1_copy.txt  file1.txt  file2.txt  file3.txt  file4.txt  file5.txt
```

- **Rename 'file2.txt 'to 'file2_renamed.txt'**

Commands:

rishita@kali:~/rishita/file_commands_lab$ mv file2.txt file2_renamed.txt

rishita@kali:~/rishita/file_commands_lab$ ls

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ mv file2.txt file2_renamed.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls
file1_copy.txt  file1.txt  file2_renamed.txt  file3.txt  file4.txt  file5.txt
```

- **Delete 'file3.txt'.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ rm file3.txt

rishita@kali:~/rishita/file_commands_lab$ ls

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ rm file3.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls
file1_copy.txt  file1.txt  file2_renamed.txt  file4.txt  file5.txt
```

- **Change to the 'backup directory and list the files:**

Commands:

rishita@kali:~/rishita/file_commands_lab$ mkdir backup

rishita@kali:~/rishita/file_commands_lab$ mv file1_copy.txt file2_renamed.txt backup

rishita@kali:~/rishita/file_commands_lab$ ls backup/

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ mkdir backup

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ mv file1_copy.txt file2_renamed.txt backup

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls backup/
file1_copy.txt  file2_renamed.txt
```

Question 17:

<u>Create a Compressed Archive:</u>

- **Go back to the 'file_commands_lab' directory and create a compressed archive of the backup directory.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ tar -czvf backup.tar.gz backup

rishita@kali:~/rishita/file_commands_lab$ ls -l

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ tar -czvf backup.tar.gz backup
backup/
backup/file2_renamed.txt
backup/file1_copy.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls -l
total 12
drwxrwxr-x 2 rishita rishita 4096 Mar  8 16:16 backup
-rw-rw-r-- 1 rishita rishita  211 Mar  8 16:17 backup.tar.gz
-rw-rw-r-- 1 rishita rishita   15 Mar  8 16:11 file1.txt
-rw-rw-r-- 1 rishita rishita    0 Mar  8 16:07 file4.txt
-rw-rw-r-- 1 rishita rishita    0 Mar  8 16:07 file5.txt
```

Question 18:

<u>Extract the Archive:</u>

- **Create a new directory named 'extracted_backup' and extract the contents of 'backup.tar.gz' into it:**

Commands:

rishita@kali:~/rishita/file_commands_lab$ mkdir extracted_backup

rishita@kali:~/rishita/file_commands_lab$ tar -xzvf backup.tar.gz -C extracted_backup

rishita@kali:~/rishita/file_commands_lab$ ls -l extracted_backup/

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ mkdir extracted_backup

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ tar -xzvf backup.tar.gz -C extracted_backup
backup/
backup/file2_renamed.txt
backup/file1_copy.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls -l extracted_backup/
total 4
drwxrwxr-x 2 rishita rishita 4096 Mar  8 16:16 backup
```

Question 19:

<u>Verify Extraction:</u>

- **List the contents of the 'extracted_backup' directory to verify that the files were extracted correctly.**

Commands:

rishita@kali:~/rishita/file_commands_lab$ ls -l extracted_backup/

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ ls -l extracted_backup/
total 4
drwxrwxr-x 2 rishita rishita 4096 Mar  8 16:16 backup
```

Question 20:

Create a text file named "count.txt" and write the 10 line sentences and use the command to count the word, lines, character and also use command for highlighting a patten in this file.

Commands:

rishita@kali:~/rishita/file_commands_lab$ touch count.txt

rishita@kali:~/rishita/file_commands_lab$ open count.txt

rishita@kali:~/rishita/file_commands_lab$ wc count.txt

rishita@kali:~/rishita/file_commands_lab$ wc -l count.txt

rishita@kali:~/rishita/file_commands_lab$ wc -w count.txt

rishita@kali:~/rishita/file_commands_lab$ wc -m count.txt

rishita@kali:~/rishita/file_commands_lab$ grep --color=auto Linux count.txt

```
┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ touch count.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ open count.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ wc count.txt
  7  36 244 count.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ wc -l count.txt
7 count.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ wc -w count.txt
36 count.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ wc -m count.txt
236 count.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ grep --color=auto rishita count.txt

┌──(rishita㉿kali)-[~/rishita/file_commands_lab]
└─$ grep --color=auto date count.txt
◆ Save the report with date and time.
```